# Nitrogen-Vacancy Center Experiments

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## ABSTRACT

This report investigates the use of Nitrogen-Vacancy (NV) centers in diamond as a platform for quantum computation. We explore the properties of NV centers, which enable long coherence times at room temperature and can be manipulated optically and via microwave pulses. Several experiments, including continuous wave spectroscopy, Rabi oscillations, spin echo, and $T_2$ relaxation, are conducted to demonstrate key quantum phenomena such as state manipulation, coherence, and decoherence in NV centers. Additionally, the paper outlines the application of the Deutsch-Josza algorithm to showcase quantum computational advantage.

## I. INTRODUCTION

Nitrogen-vacancy (NV) centers in diamond are point defects in the diamond lattice, where a nitrogen atom substitutes for a carbon atom adjacent to a lattice vacancy. These defects possess unique quantum properties, making them a promising platform for quantum computing, quantum sensing, and quantum communication. In the context of quantum computing, NV centers are particularly valued for their optically addressable electron spin states. These states can be initialized, manipulated, and read out with long coherence times at room temperature, using a combination of optical and microwave techniques [1]. NV centers fulfill four out of five of DiVincenzo's criteria [2] for quantum computing; however, scaling to a large number of qubits remains a significant challenge for researchers. In this report, we will demonstrate effective quantum computation using a single-qubit system based on NV centers.

The NV center involves six electrons in total: five from the nitrogen atom and surrounding carbon atoms near the vacancy, plus one additional electron from external sources. The spin state analysis focuses on two electrons that occupy degenerate non-bonding orbitals near the Fermi level, which are split by spin-spin and spin-orbit interactions. The energy level structure of these two electrons in the NV center is displayed in Figure 1a. The NV center features a ground-state triplet ($^3A$), an excited-state triplet ($^3E$), and two intermediate-state singlets ($^1A$ and $^1E$). The triplet states include $m_s = \pm 1$ spin states, where the two electron spins are aligned, as well as an $m_s = 0$ spin state, in which the electron spins are antiparallel. Without an external magnetic field, the $m_s = \pm 1$ and $m_s = 0$ states are split due to spin-spin interactions, and the $m_s = \pm 1$ states are degenerate. Once an external magnetic field is applied, we observe a Zeeman splitting between the $m_s = \pm 1$ states. This splitting is demonstrated in Figure 1b.

We now consider the excitation and relaxation process of the $m_s = 0$ and $m_s = \pm 1$ states according to the energy level structure and how it relates to the qubit construction of the NV center. First, the 532 nm green laser excites the ground-state electrons. We can consider many pairs of electrons because our apparatus consists of many NV centers, the states of which are manipulated simultaneously and identically. Most of the $m_s = 0$ electrons relax back to the ground state by emitting 637–600 nm red radiation, while most of the $m_s = \pm 1$ electrons transition through the non-radiative path via the singlet states. This relaxation path concludes in the $m_s = 0$ ground state. Over time, the population of $m_s = 0$ electrons increases while the population of $m_s = \pm 1$ electrons decreases. If we construct the computational basis of our qubit as $m_s = 0 \to |0\rangle$ and $m_s = \pm 1 \to |1\rangle$, this process initializes our qubit in the $|0\rangle$ state.

Since the $m_s = 0$ electrons have a greater probability of relaxing to the ground state through a radiative transition compared to the $m_s = \pm 1$, the fluorescence intensity of the two states, which differs by approximately 20–40%, can be used to distinguish between them. Therefore, exciting the electrons and measuring the fluorescence during the relaxation process serves as our qubit readout procedure.

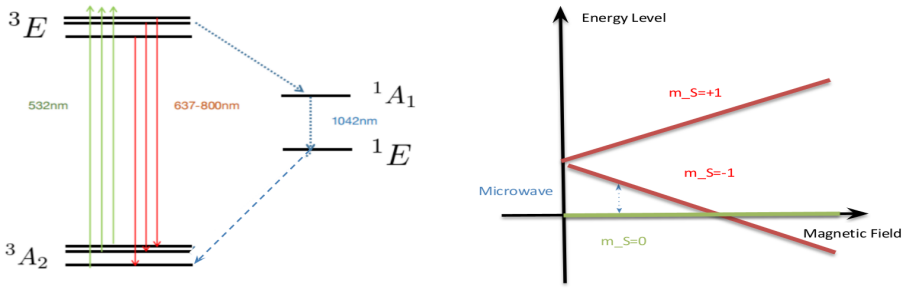
We now explore how we can coherently our spin states of the NV center. Due to the Zeeman effect [3], applying an external static magnetic field $\boldsymbol{B_0} = B_z\hat{\boldsymbol{z}}$ to a magnetic moment $\boldsymbol{\mu}$ splits the energy eigenstates by $\Delta E = \gamma\hbar B_z$. Rabi determined that if we apply a transverse oscillating magnetic field $\boldsymbol{B_1}$, with a frequency $\omega_0 = \gamma B_0$, the state of the moment will oscillate between the two energy eigenstates. For electron spin states, $w_0$ is a microwave resonant frequency. The magnitude of the probability of transition from $|0\rangle$ to $|1\rangle$ is given by:

$$P_{|0\rangle\to|1\rangle}(t) = (\frac{\omega_1}{\Omega})^2 \sin^2(\frac{\Omega t}{2}), \tag{1}$$

where,

$$\Omega = \sqrt{(\omega - \omega_0)^2 + \omega_1^2}, \tag{2}$$

where $\omega_1 = \gamma B_1$. Equation (3) defines the resonance condition of $\omega = \omega_0$ for Nuclear Magnetic Resonance [4]. When the resonance condition is met, applying a pulse for a time duration of $\Delta t = \frac{\pi}{2\omega_1}$ corresponds to a $\frac{\pi}{2}$-pulse, which rotates the magnetization around the X-axis by $\frac{\pi}{2}$. This $\frac{\pi}{2}$-pulse corresponds to a Hadamard gate which




(a) (b)

FIG. 1: 1a) Energy level structure for the NV center. 2b) Zeeman splitting due to external magnetic field of the triplet states

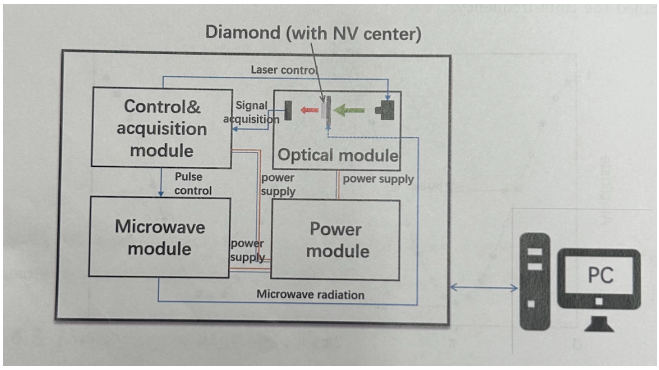



FIG. 2: Apparatus Diagram

transforms $|0\rangle \rightarrow \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ Similarly, a pulse of time duration $\Delta t = \frac{\pi}{\omega_1}$ corresponds to $\pi$ pulse, which rotates the magnetization around the X-axis by $\pi$. This $\pi$ pulse corresponds to an $X$ or $NOT$ gate.

## II. METHODOLOGY

This report investigates several experiments that demonstrate quantum computing with an NV center. The apparatus of our NV center and the methodology of the experiments will be described in the following section.

### A. Apparatus

The diagram of the apparatus for our experimentation is shown in Figure 2. The optical module includes a pulse laser generator, optical path system, microwave radiation, photodetector, and the NV center in diamond. The optical path system focuses a green laser of 520 nm on the diamond, and the NV center emits red fluorescence, which is focused into the photodetector. This process is used for qubit initialization and readout. The photodetector converts the light signal into an electrical signal, which is sent to the control and acquisition module. The microwave module generates a microwave signal of a specific frequency, modulated into pulse form, to manipulate the NV center spin state. The control and acquisition module is responsible for pulse control and signal acquisition. Pulse control generates transistor-transistor logic (TTL) signals, which are transmitted to the pulse generator and microwave module. Signal acquisition refers to the conversion of electrical signals transmitted from the photodetector into digital information, which is displayed after data processing.

### B. Continuous Wave Experiment

The purpose of the continuous wave experiment is to measure the continuous wave spectrum of the NV center. As previously discussed, applying a microwave frequency that resonates with the energy level difference induces state transitions from $m_s = 0$ to $m_s = \pm 1$. According to Figure 1b, there are two resonant frequencies, denoted as $\omega_1$ and $\omega_{-1}$, which correspond to state transitions from $m_s = 0$ to $m_s = 1$ and $m_s = -1$, respectively. To observe these transitions, we expect to see dips in fluorescence intensity at $\omega_1$ and $\omega_{-1}$. Furthermore, as the magnetic field increases, the difference between the two resonant frequencies should also increase, since the transition energy from $m_s = 0$ to $m_s = 1$ deviates from the transition energy from $m_s = 0$ to $m_s = -1$.

By constructing this plot, we can determine the values of $\omega_1$ and $\omega_{-1}$, which will be essential for the rest of the experiments. To construct this plot, we program our pulse generator to emit microwave pulses at continuously varying frequencies, while the acquisition module displays the fluorescence intensity.

### C. Rabi Oscillation Experiment

Rabi oscillations are the oscillations at resonant frequencies from $|0\rangle$ to $|1\rangle$, as described in the Introduction. This experiment serves to observe these Rabi oscillations

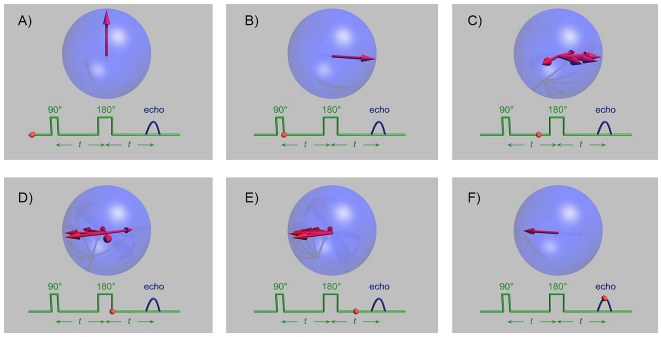



FIG. 3: Spin echo sequence

using our NV center apparatus. We achieve this by applying microwave pulses of varying lengths that oscillate at the resonant frequencies $\omega_1$ and $\omega_{-1}$, which were determined in our Continuous Wave experiment. Before each pulse, we initialize the electrons to $m_s = 0$ using prolonged exposure to a 532 nm laser. After the pulse is applied, we read out and plot the resulting fluorescence intensity as a function of pulse width. We expect to observe oscillations in the form of a decaying $\sin^2$ function, similar to Equation 1, but decaying due to loss of coherence. We can repeat this procedure for both resonant frequencies to obtain two Rabi oscillation plots. Again, we program the pulses directly into our pulse generator, and the fluorescence intensity after each pulse is acquired by the acquisition module.

### D. Spin Echo Experiment

The Spin Echo experiment applies the principles of Rabi oscillations and Nuclear Magnetic Resonance to determine characteristic relaxation times or decoherence times of a particular quantum system. The echo experiments involve the process of decoupling spin coherent signals by applying decoupling pulses. The spin echo sequence consists of a $\frac{\pi}{2}$-pulse followed by a $\pi$-pulse after a time delay of $t_1$. Only the $\omega_1$ transition frequency was used for this experiment. The pulse widths were determined from our Rabi oscillation plots, where the $\frac{\pi}{2}$ pulse width corresponds to one-quarter of the period, and the $\pi$ pulse width corresponds to one-half of the period. Using these pulse widths and resonant frequencies, the entire pulse sequence was input into the pulse generator.

If we consider the Bloch sphere representation of our qubit, the $m_s = 0$ and $m_s = \pm 1$ states can be visualized as vectors pointing along the positive and negative $z$-axis of the Bloch sphere, respectively. Figure 3 illustrates the effect of each step in the spin echo sequence on the state, as well as the emergent echo that results.

A physical description of each time step $A$–$F$ of the spin echo sequence depicted in Figure 3 is provided in the following outline:

A. Before the $\frac{\pi}{2}$ pulse, the electrons are initialized to the ground state, represented as a vector along the positive $z$-axis of the Bloch sphere.

B. Immediately after the $\frac{\pi}{2}$ pulse, the net magnetic moment is flipped to the $+\hat{\boldsymbol{x}}$-direction in the $XY$-plane of the Bloch sphere.

C. Before the $\pi$ pulse, the electron pairs comprising our qubit undergo free evolution, precessing around the $z$-axis. Due to inhomogeneities in the magnetic field, some pairs precess faster than others, leading to dephasing.

D. Just after the $\pi$ pulse, the electrons that lagged during precession now lead, and those that led are now lagging.

E. The faster electrons slow down while the slower electrons catch up, leading to rephasing of the net magnetic moment.

F. Complete refocusing occurs at $t = 2t_1$, where the peak echo signal corresponds to all electron states occupying the same coherent state in the $XY$-plane, effectively canceling the effects of magnetic field inhomogeneities.

Specific to our experiment, in order to read out the qubit state, we must apply another $\frac{\pi}{2}$ pulse after the $\pi$ pulse to flip the refocused state into our computational basis, followed by an immediate measurement. By varying the time continuously after the $\pi$ pulse within the range $t \in [t_1 - \Delta t, t_1 + \Delta t]$ and performing measurements after each sequence, we should expect to observe an echo-like plot as a function of time, centered at $t_1$, with $\Delta t$ capturing the full echo.

### E. $T_2$ Experiments

The $T_2$ experiment is used to determine the transverse decoherence time of our NV center system. Similar to our spin echo pulse sequence, to determine $T_2$, we apply a $\frac{\pi}{2}$-pulse followed by a $\pi$-pulse after a time $t_1$. After waiting another $t_1$, we should observe complete refocusing in the form of the peak of an echo. The $T_2$ relaxation time corresponds to the decay in the peak of that echo as we increase $t_1$. Therefore, by continuously varying $t_1$, applying a final $\frac{\pi}{2}$-pulse at $2t_1$, and performing a readout, we can fit our measurements to a curve of the form:

$$A(t) = Ae^{(-t/T_2)^a} + B, \tag{3}$$

and $T_2$ can be calculated.

### F. Deutsch-Josza

The Deutsch-Jozsa (DJ) algorithm is the simplest quantum algorithm that demonstrates a computational

advantage, requiring only one qubit for measurement and one additional auxiliary qubit. In our experiments so far, we have been using only the transition from $m_s = 0$ to either $m_s = \pm 1$ separately. To implement the DJ algorithm with our NV center, we can utilize both transitions simultaneously.

Consider a function $f : \{0, 1\} \to \{0, 1\}$. Since the domain and range of this function consist of just one bit, there are four possible functions: $f_1(x) = 0$, $f_2(x) = 1$, $f_3(x) = x$, and $f_4(x) = 1 - x$. Notice that $f_1$ and $f_2$ are constant functions, while $f_3$ and $f_4$ are balanced. The DJ algorithm, when given black-box access to an unknown function $f$, can determine whether $f$ is balanced or constant with only one query. The best possible classical algorithm would require two queries on both inputs. Figure 4 presents the quantum circuit for the DJ algorithm.

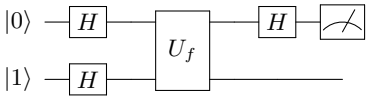



FIG. 4: Quantum circuit diagram of the Deutsch-Jozsa algorithm

By analyzing the circuit, we can conclude that if $f$ is a constant function ($f(0) = f(1)$), the final state becomes:

$$|\psi\rangle = \pm\,|0\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right). \tag{4}$$

If $f$ is a balanced function ($f(0) \neq f(1)$), the state becomes:

$$|\psi\rangle = \pm\,|1\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right). \tag{5}$$

Therefore, by measuring our qubit, we can determine whether $f$ is constant or balanced with just one query. Using our NV center system, we will encode $m_s = 0$ and $m_s = -1$ as our qubit states, with the $m_s = 1$ state serving as our auxiliary level. The DJ algorithm requires that we can apply Hadamard gates, which correspond to $\frac{\pi}{2}$ pulses at either microwave frequency $\omega_{-1}$ or $\omega_1$, measure the qubit, and apply $U_f$, which operates as follows:

$$U_f\,|x, 0\rangle = |x, f(x)\rangle\,, \quad U_f\,|x, 1\rangle = |x, 1 - f(x)\rangle\,. \tag{6}$$

For each of the four possible functions, we need a different pulse sequence to implement $U_f$. Figure 5 shows the pulse sequences for each $f$, as well as the matrix representations of $U_f$.

The structure of each pulse sequence is the same as our spin echo experiment, but with $2\pi$ pulses inserted according to our specific $f$. The key point here is that when our qubit is in the Hadamard basis, applying $2\pi$ pulse on our auxillary bit generates a $\pi$ phase shift on our qubit around the $z$ axis. This should be done for our

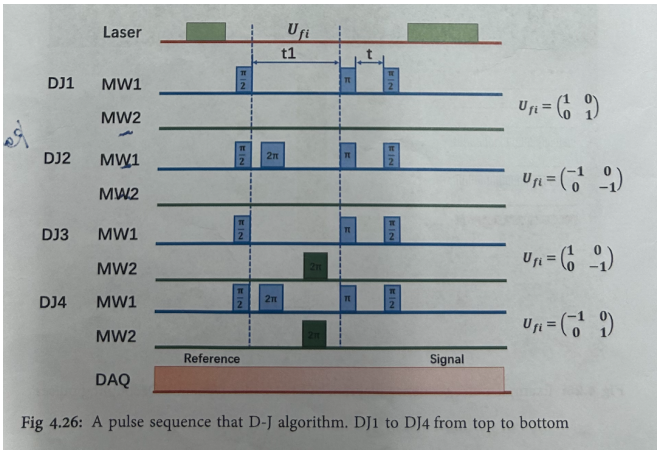



FIG. 5: Pulse sequence for the Deutsch-Josza algorithm implementation

$U_f$ implementations for only balanced functions. Therefore, after applying $U_f$ for constant functions we should expect the typical spin echo plot and for balanced functions we should expect a reverse spin echo. This is how we determine if $f$ is constant or balanced, completing the goal of the DJ algorithm.

## III. RESULTS

The first experiment conducted was the continuous wave experiment, and the results are shown in Figure 6a). As we vary the microwave frequency, we observe two valleys corresponding to the transition frequencies from $m_s = 0$ to $m_s = \pm 1$. From the plot, we determine that $\omega_{-1} = 2844$ MHz and $\omega_1 = 2895$ MHz. These frequencies will be used for the remaining experiments.

The next experiment conducted was the Rabi oscillation experiment. Using $\omega_{-1}$ and $\omega_1$, we input these frequencies into our microwave module, which stimulates transitions from the $m_s = 0$ state to the $m_s = \pm 1$ states. Figure 6b) shows the Rabi oscillations for the two transitions, where MW1 corresponds to the transition to $m_s = 1$. We observe that the oscillations decay due to decoherence, caused by the inhomogeneities of the magnetic field. The difference in oscillation frequencies results from the slight deviation of our inputted frequency from the actual transition frequency, as described by Equation 2, since the $\omega_1$ frequencies are similar. From these oscillations, we calculate the time duration for a $\frac{\pi}{2}$-pulse and $\pi$-pulse, corresponding to $T/2$ and $T/4$ for both MW1 and MW2. We calculate that for MW1: 100 ns $\to \pi$ and 50 ns $\to \frac{\pi}{2}$, and for MW2: 60 ns $\to \pi$ and 30 ns $\to \frac{\pi}{2}$.

For our echo experiments, we use MW1 and the corresponding pulse durations to apply the spin echo sequence, and the resulting spin echo sequence is shown in Figure 6c). Our inputted $t_1$ time was 1000 ns, and we observe the peak of our spin echo at precisely 1000 ns, counting

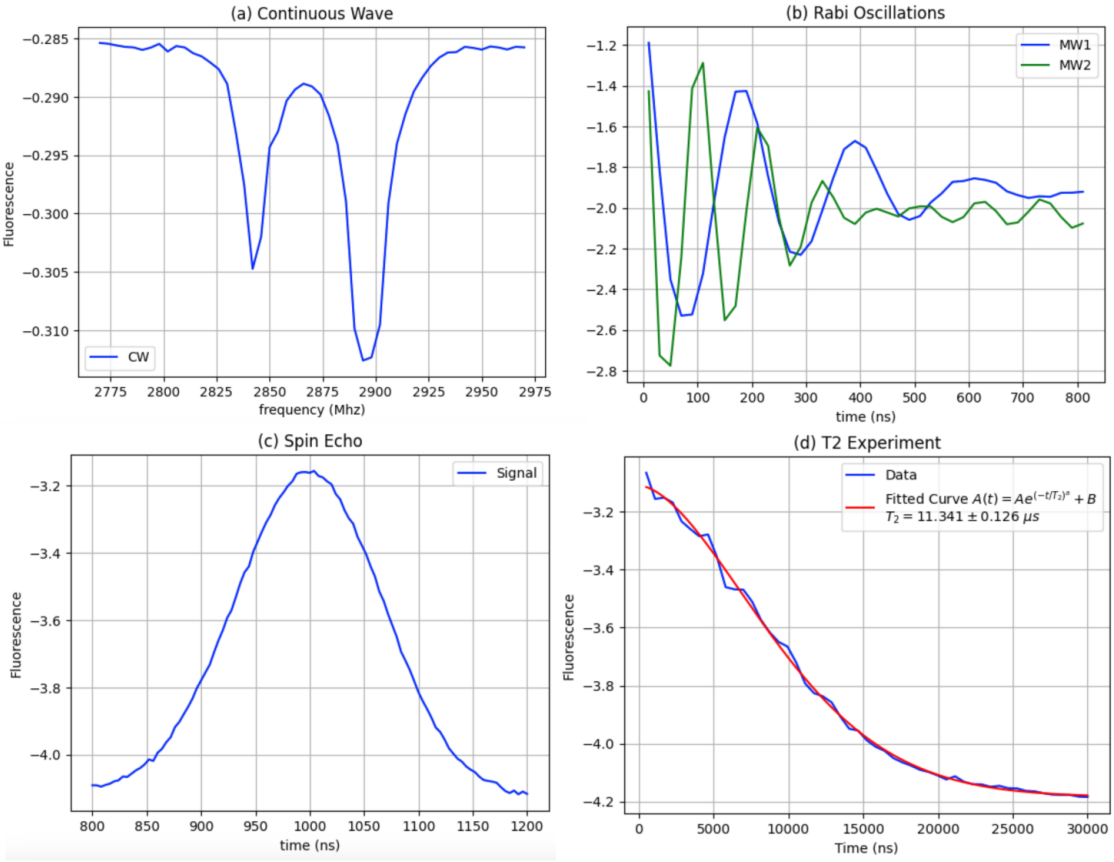



FIG. 6: Results for a) Continuous Wave b) Rabi Oscillations c) Spin Echo d) T2 Experiments

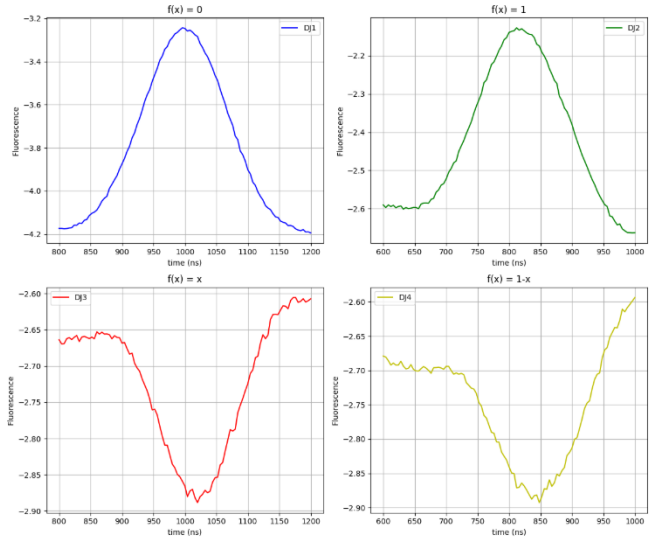



FIG. 7: output of DJ algorithm for each of our four $U_f s$

from the $\pi$ pulse, with a width of around 400 ns.

For the T2 experiment, we again use MW1 for our transition frequency and use the corresponding pulse widths for the pulse sequence. The decaying curve due to decoherence is shown in Figure 6d). We fit the curve to the form in Equation 3 and calculate:

$$T_2 = 11.341 \pm 0.126\ \mu\text{s}. \tag{7}$$

Finally, for the Deutsch-Josza experiment, we observe the spin echoes or reverse spin echoes for our four 1-bit functions, shown in Figure 7. We conclude that $f_1$ and $f_2$ are constant functions because they exhibit the typical spin echo plot. Similarly, we conclude that $f_3$ and $f_4$ are balanced functions because they show the reverse spin echo plot. The DJ algorithm performs as expected.

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